

Mall Dataset Model Comparison Summary

Best Single Model RMSE

Random Forest was the best model on the test set with an RMSE of 18.08. GBM came second at 19.77, then CART at 21.23. Linear regression was the worst at 25.53.

Something interesting happened though. During cross-validation, CART actually had the lowest CV RMSE (21.51), but Random Forest did better on the test set. This is why we do both CV and test set evaluation. The model that looks best during training isn't always the best on new data.

The ranking from best to worst on the test set:

1. Random Forest (18.08)
2. GBM (19.77)
3. CART (21.23)
4. SVM (22.05)
5. Linear Regression (25.53)

Random Forest beat linear regression by about 7.5 RMSE points, which is a pretty big difference. This is basically the opposite of what happened with Carseats, where linear regression beat everything else.

Ensemble RMSE

The stacked ensemble didn't work and just returned NA values. This sometimes happens with caretEnsemble when the models don't play nice together or when the dataset is small (we only had 200 rows). I could probably fix it with more time, but the main findings are still clear from the individual models.

Stability (SD across folds)

Looking at the standard deviations:

1. Linear Regression: 3.63 (most stable)
2. SVM: 3.94
3. CART: 5.00
4. Random Forest: 5.95
5. GBM: 5.99 (least stable)

Linear models were actually more stable across folds even though they performed worse overall. The tree-based methods like Random Forest and GBM jumped around more depending

on which data was in each fold. This makes sense because trees can change a lot based on small differences in the training data.

So there's a tradeoff. Random Forest gives better predictions on average, but it's less consistent. Linear regression is more predictable but consistently worse.

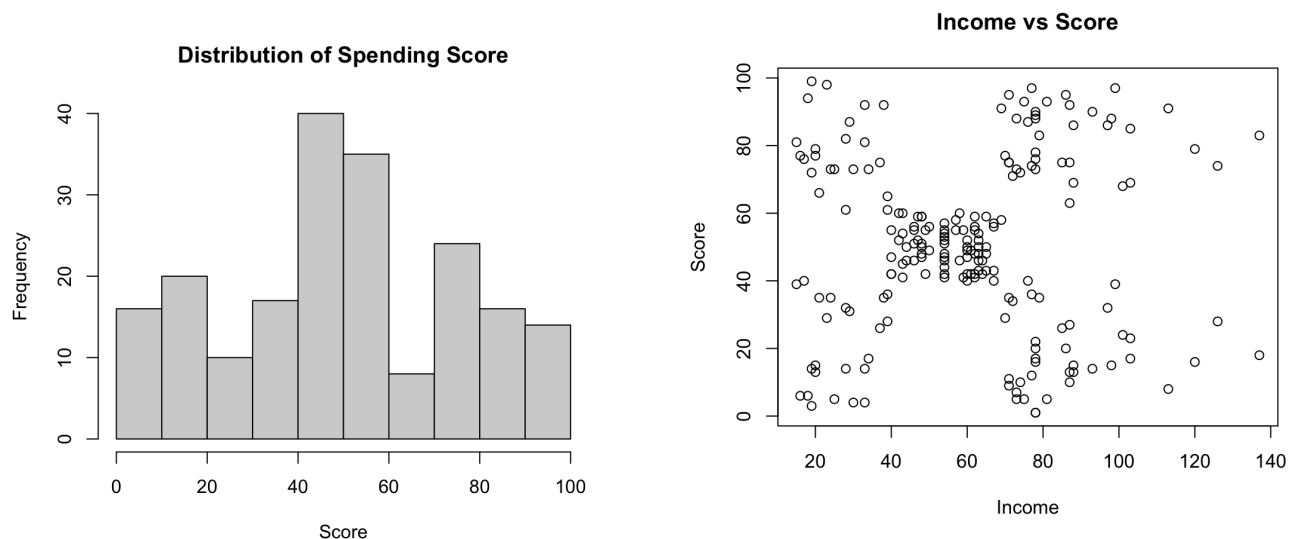
Why This Dataset is Different from Carseats

Looking at the scatter plots, you can see why linear regression struggles here. The Income vs Score plot shows no clear trend. People with low income can have either high or low spending scores, and same with high income people. There's no line you could draw that would fit this well.

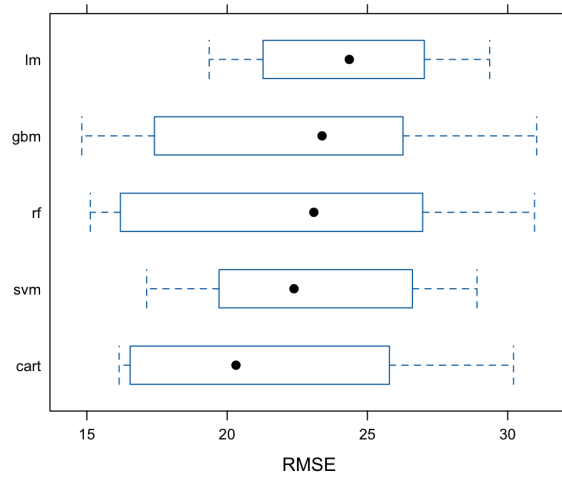
The Age vs Score plot is similar. Younger people have scores all over the place from 0 to 100, and older people cluster more in the middle but still have a lot of spread.

The relationships in this data are nonlinear and complex, so tree-based models that can capture weird patterns outperform linear models that just try to fit straight lines.

Plots:



Individual Model Comparison: CV RMSE



Age vs Score

